tension. For example, a highly soluble substance like sodium chloride (*NaCl*) when dissolved in water (H_20) increases the surface tension of water. But the sparingly soluble substance like phenol or soap solution when mixed in water decreases the surface tension of water.

(3) ***Electrification*** affects the surface tension. When a liquid is electrified, surface tension decreases. Since external force acts on the liquid surface due to electrification, area of the liquid surface increases which acts against the contraction phenomenon of the surface tension. Hence, it decreases.

(4) ***Temperature*** plays a very crucial role in altering the surface tension of a liquid. Obviously, the surface tension decreases linearly with the rise of temperature. For a small range of temperature, the surface tension at T_t at t °C is $T_t = T_0\ (1 - \alpha\, t)$

Where, T_0 is the surface tension at temperature 0°C and α is the temperature coefficient of surface tension. It is to be noted that at the critical temperature, the surface tension is zero as the interface between liquid and vapour disappear. For example, the critical temperature of water is 374°C. Therefore, the surface tension of water is zero at that temperature. van der Wall suggested the important relation between the surface tension and the critical temperature as

$$T_t = T_0 \left(1 - \frac{t}{t_c}\right)^{\frac{3}{2}}$$

Generalizing the above relation, we get

$$T_t = T_0 \left(1 - \frac{t}{t_c}\right)^{n}$$

which gives more accurate value. Here, n varies for different liquids and t and t_c denote the temperature and critical temperature in absolute scale (Kelvin scale) respectively.

7.5.3 Surface energy (S.E.) and surface tension (S.T.)

Surface Energy

Consider a sample of liquid in a container. A molecule inside the liquid is being pulled in all direction by other molecules that surround it. However, near the surface, a molecule is pulled down only by the molecules below it and there is a net downward force. As a result, the entire surface of the liquid is being pulled inward. The liquid surface thus tends to have the least surface area. To increase the surface area, some molecules are brought from the interior to the surface. For this reason, work has to be done against the forces of attraction. The amount of work done is stored as potential energy. Thus, the molecules lying on the surface possess greater potential energy than other molecules. This excess energy per unit area of the free surface of the liquid is called '*surface energy*'. In other words, the work done in increasing the surface area per unit area of the liquid against the surface tension force is called the surface energy of the liquid.

$$\text{Surface energy} = \frac{\text{work done in increasing the surface area}}{\text{increase in surface area}}$$

$$= \frac{W}{\Delta A} \qquad (7.26)$$

It is expressed in J m^{-2} or N m^{-1}.

Surface tension

The surface tension of a liquid is defined as the force per unit length of the liquid or the energy per unit area of the surface of a liquid

$$T = \frac{F}{l} \qquad (7.27)$$

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